Labwork 1

I. Matlab introduction

1. Vector

Exercise 1:

Define a row vector x, one row, five column with value = (1, 2, 3, 4, 5); find the value x(3)+x(2)=?

Exercise 2: Define a **column** vector x = (1, 2, 3, 4, 5); Transpose this vector?

Exercise 3: Calculate: a = 1 + 2+ … + 100 using the function *sum*

Exercise 4: Define the symbolic function x + y using the function *syms*. Find the value of f at x = 1 and y = 2

Exercise 5: Use the function *symsum* to calculate

a) $\sum_{k=1}^{n}\left(\frac{1}{k}-\frac{1}{1+k}\right).$

b) $\sum_{n=1}^{\infty}\frac{1}{n^2}.$

2. Matrix

Exercise 1

a. Define matrix $A=\begin{bmatrix}0 & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}$ by two approaches: Define element by element and used the function *zeros*
b. Change the diagonal of A to 1 without define the matrix again using *diag*
c. Define B = A transposed
d. Calculate A+B, A-B, A*B and A.*B. Give remark about the results of A*B and A.*B
e. Reduce the size of A to 2x2 by removing row 3 and column 3 of the original matrix. Do the same for B but remove row 1 and column 1. *Hint: use the operator* "*:*"
f. What's the difference A(1:3) and A(1:3,1) or A(:,1)?

3. Function

Exercise 1

Write a function that calculates the mean of a vector.

Application: vector x = 1:3:100

Exercise 2

Write a function that solves the quadratic equation $ax^2 + bx + c = 0$.

Application: equation $x^2 + 2x - 3 = 0$

4. if else structure

Exercise 1

Write brief scripts to evaluate the following functions.

m = n + 1 with n >1

m = n - 1 with n <1

Application:

a) n = 5

b) n = -5

Exercise 2

Write brief scripts to evaluate the following functions.

y = 4x with 0 < x <10

y = 10x with 10 < x <40

y = x with x > 40 and x < 0

Application:

a) x = 5

b) x = -5

c) x = 20

Exercise 3

Electricity bill: In Vietnam, the electricity price varies with the number kWh, e.g the first 50 kWh will be cheaper than the next 50 kWh. In particular:

- From 0 to 50 kWh: 1.484 VND / kWh
- From 51 to 100 kWh: 1.533 VND / kWh
- From 101 to 200 kWh: 1.786 VND / kWh
- From 201 to 300 kWh: 2.242 VND / kWh
- From 301 to 400 kWh: 2.503 VND / kWh
- From 400 kWh: 2.587 VND / kWh

Write a script to calculate the price of electricity with input is the number of kWh

5. For loop

Exercise 1:

Write a script to calculate these sums:

a. $1^2+2^2+3^2+...+1000^2$

b. 20!

c. $1-\frac{1}{3}+\frac{1}{5}-\frac{1}{7}+\frac{1}{9}-...-\frac{1}{1003}$

d. Prove that: $\frac{1}{1^2\cdot 3^2}+\frac{1}{3^2\cdot 5^2}+\frac{1}{5^2\cdot 7^2}+...=\frac{\pi^2-8}{16}$

Exercise 2:

Write a script to calculate:

a. Combination of a set ($C_n^k=\frac{n!}{k!(n-k)!}$)

b. Permutation of a set ($A_n^k = \frac{n!}{(n-k)!}$)

Hint: write a function to calculate the factorial of a number first, then use the function to write the required script)

6. While loop

Exercise 1:

a) Write code with *while loop* that makes the computer count number from 1 to N (N: integer), for example from 1 to 5, there are 5 number.

b) Write code with *while loop* that makes the computer count only the odd number from 1 to N (N: integer), for example from 1 to 5, there are 3 odd number.

Exercise 2:

Calculate the sum S of elements a =√2i-1, i=1, 2, ..., until the sum will exceed 20 using *while loop*.

Exercise 3:

Doubling time of an investment: write a script to calculate hold long it will take for our bank investment to double. Suppose that the interest rate is 10% per year and all the interest is added to the initial investment.

7. Graphic

Exercise 1:

a. Draw the graph of a linear function y = x+1 in the range of x = -5 to x = 5. *Hint:* consult help to know how to use the function *linspace* and *plot*.

b. Change the color and the pattern of the line using *plot* function (*Hint: the option for colors and patterns can be found in the help sections*)

c. Plot the line y = -x + 2 on the same plot.

d. Add title, legend and labels for each axis.